Adaptive Contraction Trees with Exact Support Certificates

A sharper exponential upper bound after reviewing twenty SAT refinement packets

Research prepared for Vladimir Reshetnikov

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Abstract

We review the two primary manuscripts, fifteen placed refinement packets, and five newer incoming archives in ProveIt at `fa6e904d109b738154db7fe41e7f2777e6d24aee`. Under their common normalized Boolean tensor-contraction interface, we obtain time

$$(s+\mathcal{L})^{O(1)}2^{(30.891018920366\ldots+o(1))s/\sqrt{\lceil \log_2 s\rceil}}$$

and space with coefficient $30.753912832055\ldots$ on the same exponential scale. The improvement comes from selecting among three contraction trees for each exact mode-weight allocation. An elementary support-averaging lemma reduces its uniform analysis to finite integer certificates: 9,261 mask triples at nine groups and 35,937 at thirteen groups. A strict rational $9 \to 13 \to 7$ schedule pays for every recursively evaluated child entry.

The strongest submitted comparable coefficient was $35.354588058401\ldots$; combining its ingredients with an existing boundary-retreat technique already gives $35.063598846170\ldots$. The new bound improves both. It is a coefficient improvement in an exponential upper bound, with subexponential space. It does not improve the logarithmic exponent order for unrestricted SAT, establish the conditional $n/\log n$-type targets, or independently certify the upstream ETH claim. We also prove the exact full-table boundary width of a complete directed mode graph, isolating why adaptive trees alone do not remove the square-root logarithm in this representation.

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1 Outcome and interpretation

The principal result is a new, rigorously checkable coefficient certificate for the same normalized contraction problem used by the latest packets. Put $\lambda = \lceil \log_2 s \rceil$. Our exact constants are

$$C_T = \frac{\pi}{\ln 2}\left(\sqrt{\frac{128000}{5103}} + \sqrt{\frac{134144}{41067}}\right) = 30.891018920366092\ldots, \tag{1}$$

$$C_S = \frac{\pi}{\ln 2}\sqrt{\frac{302080}{6561}} = 30.753912832054975\ldots. \tag{2}$$

Thus 30.892 and 30.754 are convenient eventual time and space coefficients. The variable s counts normalized Boolean wires; it is not the number of SAT variables. The conversion of a wire coefficient into a SAT-variable coefficient requires the reduction's constants, which are not supplied by the reviewed numerical comparisons.

Comparable generic contraction certificate	Time	Space
Best submitted cached tree packet [10]	35.354588	34.274803
Existing ingredients plus boundary retreat, Section 2	35.063599	33.992701
This report: three-tree selection and $9 \to 13 \to 7$ schedule	**30.891019**	**30.753913**

Every entry multiplies $s/\sqrt{\lambda}$ in a base-two exponent, with a vanishing additive coefficient and polynomial factors. These are upper-bound comparisons, not observed runtimes. The new time coefficient is about 12.6% below the submitted tree certificate and 11.9% below the stronger displayed composition. In particular, if U_{old} and U_{new} denote the corresponding explicit asymptotic upper-bound expressions, then

$$\log_2 \frac{U_{\mathrm{old}}}{U_{\mathrm{new}}} = (4.172579925804\ldots + o(1))\frac{s}{\sqrt{\lambda}}$$

for the 35.063599 comparison. This does not give a lower bound on the runtime of an older algorithm.

Scope of the request. The new generic normalized certificate is smaller than all displayed comparable generic certificates in the reviewed packets. Several low-degree specializations have smaller numerical constants under stronger hypotheses; they remain applicable. Conditional $n/\log n$-type proposals are not proved algorithms to which this result can claim numerical dominance. The separate Claude manuscript has an unspecified big-O constant. No claim is made to strictly dominate that unknown constant, every possible composition of the corpus, or a globally optimal schedule. The useful envelope retains all established applicable branches and adds the new generic branch.

2 Review of recent commits and the complete baseline

2.1 Frozen corpus and evidence

The review uses ProveIt commit `fa6e904d109b738154db7fe41e7f2777e6d24aee`, whose latest incoming addition is dated 2026-10-27 02:31:45 UTC (October 26 in the user's Pacific time zone). The relevant sequence is:

Commit prefix	Relevant change
`a505bd768b`, `64b3202ee8`, `8afdbbea48`	First-round SAT/ETH arrivals.
`2b925d8800`	Seven first-round packets placed in SubexponentialSAT.
`3e09608991`	Upstream preprint vendored at `7ef5c0cc5c`.
`45233238ae`	Summary of the strongest first-round claims.
`3e57705cf1`, `d098ba436b`	Eight second-round arrivals.
`6d3adc4506`	Those eight packets placed in SubexponentialSAT.
`f42479bca2`, `f898a9fde4`, `fa6e904d10`	Five additional archives in `docs/incoming` at the review pin.
`450252e9d8`	Closing check: the same five packets moved into Research.

At the initial pin, the project README indexes only the fifteen placed packets. The five incoming archives are essential to the present comparison. They were safely extracted and inspected; supplied benchmark results were not treated as measurements made in this session. The source manifest records immutable paths, Git object identities, and SHA-256 hashes. The archive does not duplicate the primary manuscripts.

The closing check used repository head `dde01c4a63b822c25b8eb345299346c36f837b49`. Commit `450252e9d8` had moved the five archives into the project and restored two earlier replay scripts. All 230 placed files were independently checked against their new Git blob identities and are byte-identical to the reviewed archive members. No additional mathematical bound was introduced. At that closing head, `docs/incoming` contains only its README. The original pin and immutable archive links below remain the provenance of the mathematical review; the closing relocation record supplies the current path mapping.

2.2 Inventory of twenty refinement packets

The following table is a compressed inventory. The accompanying `audit/baseline.txt` provides source anchors and the longer comparison. "PS" means polynomial space. "Cached" permits subexponential space. Coefficients are generic normalized-wire coefficients unless explicitly qualified. All results continue to depend on their stated imported lemmas.

Packet	Strongest relevant contribution or limitation
`cauchy_mode_improvement`	Factorial modes with weights $3b + 1 - 2\,\mathrm{popcount}(b)$; coefficient gain, unchanged order.
`eth_contraction_improvements`	Sharper Hermite taper, precision and extraction; PS coefficient about 648.44.
`3sat_precision_refinement`	Precision $24MD/u \to 6MD/u$, taper minimax; $s/\log s$ contraction remains conditional.
`sat_density_improvement`	Uniform density-sensitive SAT envelope in (3).
`eth_density_refinement`	Independently obtains the intermediate-density gain; order unchanged at fixed density and in the worst case.
`3sat_refinement`	Scalar parity lift; a compact second-contraction interface is missing.
`3sat_asymptotic_analysis`	Restricted taper/rank/degree barriers and unconstructed interfaces.

Packet	Strongest relevant contribution or limitation
`eth_asymmetric_packet_balancing`	Generic PS 76.90899; stronger low-degree premise gives 31.0641.
`3sat_round2_refinement`	Generic coefficient 84.06627; actual-domain rank and alternative-manuscript audits.
`eth_balanced_packing`	Near-unit padding and thin precision gap; 94.74820.
`eth_high_moment`	Independent 94.74820 bound, plus degree-sensitive specializations.
`eth_matching_refinement`	Third independent 94.74820 bound via matching-potential packing.
`eth_packet_refinement`	Improved pair loads and additive guard; 95.66394.
`eth_recursive_refinement`	Modulo-eight mode refinement; 133.20737.
`eth_joint_cauchy`	Full-block slope-four modes; PS 319.68576, cached Strassen 225.59706. Its ingredients combine with better schedules.
`sat_adaptive_contraction`	New incoming: streamed PS 55.70948; adaptive cached 36.75031; low-degree cached 15.39193.
`sat_frontier_improvements`	New incoming: variable arity PS 63.85741, cached 38.48450, improved exact kernels.
`subexponential_sat_frontier`	New incoming: variable arity PS 63.85220, cached 37.93011, balancing and precision-peeling kernels.
`subexponential_sat_kernel_improvements`	New incoming: actual-domain Smith separation, Gray/projective kernels; no smaller uniform global order.
`subsat_contraction_tree`	New incoming: recursive block tables and exact ring bridge to fast matrix multiplication; cached 35.35459.

2.3 Established order envelope and conditional frontier

For fixed clause width k and n active variables, the strongest established density-sensitive order in the corpus is

$$(n+m)^{O_k(1)}2^{O_k\left(\frac{n}{\sqrt{\log n}}\min\{\log(2+m/n),\log\log n\}\right)}. \tag{3}$$

The density constants are uniform in the stated range [4]. Unused variables can be removed and their count restored by a power of two. The direct clause-network branch with exponent $O_k(m/\sqrt{\log(m+2)})$ is retained when useful [2, 3]. Small sizes are handled separately, so no expression involving a small or negative iterated logarithm is used literally.

At $m=\Theta(n)$ the exponent scale is $n/\sqrt{\log n}$; at $m=\Theta(n\log\log n)$ it is $n\log\log\log n/\sqrt{\log n}$; the unrestricted worst-case scale is $n\log\log n/\sqrt{\log n}$. Neither our method nor any of the five current incoming packets changes these orders. The strongest conditional proposal remains

$$2^{O(n\log\log n/\log n)} \quad \text{in general,} \qquad 2^{O(n/\log n)} \quad \text{at fixed density,}$$

subject to the unproved structured-mode summation algorithm of [5]. Its existence is not implied by compact rank encodings: the encoded group tensors can violate the degree conditions of the contraction theorem.

2.4 A stronger comparison than the best printed number

The incoming tree method uses a root alphabet of eight and a tail alphabet of twelve. Its displayed schedule deliberately leaves rational slack. An already available inward-boundary technique permits

$$u_0 = \left\lfloor \frac{\lambda}{60} - \sqrt{\lambda} \right\rfloor, \qquad u_d = \left\lfloor \frac{\lambda}{108\,6^{d-1}} \right\rfloor \quad (d \geq 1).$$

The leading total capacity is $1/36$. The negative square-root term absorbs the root's logarithmic guards; the tail has the original strict packing margins. With $\kappa = \pi\sqrt{2}/\ln 2$, this gives time

$$\max\left\{ \frac{\sqrt{798/25}}{8}\,\kappa\sqrt{60}, \quad \frac{\sqrt{14}}{8}\,\kappa\sqrt{60} + \frac{\kappa\sqrt{108}}{4}\max\left(\frac{\sqrt{1728/25}}{12}, \frac{\sqrt{22}/12}{1-1/\sqrt{6}} \right) \right\}.$$

The two terms are $35.063598846170\ldots$ and $34.758940941287\ldots$. This is a feasible compatible baseline, not a claim of global schedule optimality. Likewise, composing the earlier variable-arity schedule with streamed separator evaluation gives a PS coefficient $55.404665906611\ldots$, below the printed 55.70948. Our cached result is compared with these stronger compositions, while respecting their different memory guarantees.

3 The exact normalized interface and inherited ingredients

The root is a bipartite Boolean tensor network with at most s wires and maximum degree at most $\lceil s^{9/10} \rceil$. Its entries lie in

$$(\mathbb{Z}/2^{P_0}\mathbb{Z})[V]/(V^{\ell}), \qquad P_0, \ell \leq s.$$

The input-description length is $\mathcal{L}$. Entry routines permitted by the reviewed interface take $(s + \mathcal{L})^{O(1)} 2^{O(s^{9/10})}$ time and polynomial space, with fixed constants. The source's extension ring, immutable creation-precision tags, entrywise canonical lifts, Cauchy deformation, and Hermite recovery are retained [2, 7, 10]. This report supplies new aggregation and resource analysis, rather than an independent proof of all those ingredients.

At an active call, let M be its wire count, r its fixed group count, u its taper length, a its cumulative packet exponent, i a retained valuation, and $D = 2^{a-i}$. The combined inherited interface provides:

1. Near-unit common pair padding, with padded mass $(1 + o(1))M$ and child size at most $(2/r + o(1))M$.
2. For every directed pair of distinct groups, at most $(2 + o(1))M/(r^2 D)$ mode families of each index q, with weights $a + 1 - i + 4q - 2\,\mathrm{popcount}(q)$.
3. Sharp recovery precision $p_t \leq (6 + o(1))MD/u$, using an additive logarithmic guard to absorb inherited precision.
4. Exact group factors Z_g whose entries are recursively evaluated child contractions. These entries are not unit-cost oracles.

Allocate exact nonnegative weights b_{gh} with $\sum_{g \neq h} b_{gh} < p_t$. For each directed pair, let $\mathcal{I}_{gh}$ be the set of its mode-exponent tuples of exact weight b_{gh}. Skip an allocation if one list is empty. Write

$$h_{gh} = \log_2 |\mathcal{I}_{gh}|, \qquad \mathcal{K}(M, u) = \kappa \frac{M}{\sqrt{u}}, \qquad \kappa = \frac{\pi\sqrt{2}}{\ln 2}.$$

The corresponding finite sum is

$$\sum_{\xi_{gh}\in\mathcal{I}_{gh}} \prod_{g=1}^{r} Z_g((\xi_e)_{e\ni g}). \tag{4}$$

There are polynomially many exact-weight arrays because r is a member of the fixed set $\{7, 9, 13\}$ and all budgets are polynomial in s.

Lemma 3.1 (Inherited weighted entropy estimate)**.** *For every fixed collection of nonnegative directed weights z_{gh}, uniformly over the exact-weight arrays and active depths used below,*

$$\sum_{g\neq h} z_{gh}h_{gh} \le \left(\frac{\sqrt{\sum_{g\neq h} z_{gh}^2}}{r} + o(1)\right)\mathcal{K}(M,u). \tag{5}$$

Proof. This is the shared full-Cauchy counting argument of [8, 10]. For completeness, lowering weights to $1 + 4q - 2\,\mathrm{popcount}(q)$ gives

$$\sum_{q\ge 0} -\ln(1 - e^{-\theta(1+4q-2\,\mathrm{popcount}(q))}) = \frac{\pi^2}{24\theta} + O(\log^2(2/\theta)).$$

For pair e, use parameter θ/z_e and multiply the logarithm of its generating-function bound by z_e. Summing gives leading terms

$$\theta p_t + \frac{\pi^2}{24\theta}\frac{(2+o(1))M}{r^2 D}\sum_e z_e^2.$$

Optimization in θ gives (5). The common saddle scale is $\sqrt{u}/D$; the guard makes the error lower order uniformly. Zero weights are omitted. The joint bound does not assume a separate large-budget approximation for each individual exact list. □

Exact list sizes are computable by the standard unbounded-knapsack recurrence

$$N(j,b) = N(j-1,b) + N(j,b-w_j), \qquad N(0,0) = 1,$$

with zero values for negative budgets and $N(0,b) = 0$ for $b > 0$. Counts have polynomial bit length. The same table gives enumeration and rank/unrank operations with polynomial overhead. Thus later selections use explicit integers, not a counting oracle or floating-point decisions.

4 A uniform certificate for adaptive contraction trees

4.1 Operation supports

For disjoint vertex sets A, B, put $O = [r]\setminus(A\cup B)$. The table for A retains all pair indices between A and its complement, and sums indices internal to A. Merging the tables for A and B is a rectangular matrix multiplication. Its three independent index classes join A to O, A to B, and B to O.

Let their logarithmic dimensions be H_1, H_2, H_3. Using the existing exact coordinate/CRT bridge and the established strict matrix exponent $\omega_{\mathbb{Q}} < 12/5$ [11, 10], the operation has time exponent

$$H_1 + H_2 + H_3 - \frac{3}{5}\min(H_1,H_2,H_3) = \max_{j=1,2,3}\left(\frac{2}{5}H_j + \sum_{k\neq j} H_k\right). \tag{6}$$

The final three-block contraction uses the same formula for the triangle of block-pair indices. Its concluding trace or inner product is no larger.

For clarity about arithmetic, the bridge expands a ring entry in polynomially many integer coordinates, lifts coordinates to $[0, 2^p)$, and computes coordinate matrix products modulo sufficiently many nonexceptional odd primes. CRT reconstructs their nonnegative integer values, then ring structure constants are applied and the result reduced modulo 2^p. Each reconstruction bound has polynomial bit length. A fixed rational bilinear matrix algorithm with exponent below $12/5$ supplies constant-size identities; clearing its fixed denominators excludes only finitely many primes. Square blocking gives (6). This neither divides by a nonunit in the source ring nor assumes a field structure for it. Sequential recursion and reconstruction use a polynomial factor times input/output table storage. The finite checker tests exact regrouping; it does not implement this theoretical fast backend.

For an undirected pair $e = \{g, h\}$, set $g_e = h_{gh} + h_{hg}$. Multiplying each linear form in (6) by five gives a vector $v \in \{0, 2, 5\}^{\binom{r}{2}}$. Its entries are zero on edges internal to a part of the operation tripartition, two on its discounted class, and five on the other two classes.

A tree plan first partitions the ordered groups into three nonempty blocks. Within each block, split its ordered list at $\lfloor k/2 \rfloor$, recur on the two pieces, and merge them. Finally contract the three block tables. There are $r - 3$ binary merges and one final triangle, hence $3(r - 2)$ support vectors. Let $\mathcal{V}_j$ be the support set of plan j and define

$$F_j(g) = \frac{1}{5} \max_{v \in \mathcal{V}_j} v \cdot g. \tag{7}$$

Up to polynomial factors, $2^{F_j(g)}$ pays for all its local operations after the group-function entries have been cached.

Theorem 4.1 (Finite support-averaging certificate)**.** *Fix q tree plans on r groups. Define the exact integer*

$$Q = \max_{v_j \in \mathcal{V}_j} \|v_1 + \cdots + v_q\|_2^2. \tag{8}$$

Selecting the plan with least $F_j(g)$ gives local nonrecursive time and local storage at most a polynomial factor times

$$2^{(b+o(1))\mathcal{K}(M,u)}, \qquad b = \frac{\sqrt{2Q}}{5qr}.$$

Selection is deterministic and polynomial-time in the original size parameter. The estimate is uniform over every feasible exact-weight array.

Proof. Pointwise for every nonnegative g,

$$\begin{aligned} \min_j F_j(g) &\le \frac{1}{q} \sum_j F_j(g) \\ &= \frac{1}{5q} \max_{v_j \in \mathcal{V}_j} (v_1 + \cdots + v_q) \cdot g. \end{aligned}$$

For each tuple of support vectors, assign both orientations of an edge the weight $(v_1 + \cdots + v_q)_e/(5q)$. Its squared directed norm is $2\|v_1 + \cdots + v_q\|_2^2/(25q^2)$. Apply Lemma 3.1. The menu and all supports are fixed and finite, so the $o(1)$ is uniform over all tuples. There is no exchange of a minimum and a maximum and no assumption that vertex permutations act as arbitrary edge permutations.

For selection, compute each integer

$$P_j = \max_{v \in \mathcal{V}_j} \prod_{\{g,h\}} \left(|\mathcal{I}_{gh}|\, |\mathcal{I}_{hg}| \right)^{v_{\{g,h\}}}.$$

Comparing P_j exactly is equivalent to comparing $F_j(g)$. Their bit lengths are polynomial, and q, r are fixed.

Finally, every local table is an input or output of one of the chosen plan's matrix operations. Each such table has logarithmic size equal to a sum of two H_i's. For nonnegative H_i, expression (6) is at least every such sum. The imported bridge's storage bound therefore puts every local table below $2^{F_j(g)}$ up to a polynomial factor. Singleton caches are inputs to a first merge or the final triangle, so their storage is included. Their expensive recursive evaluation time is accounted for separately in Section 5. □

What is new here. The prior adaptive packet minimizes total crossing entropy and then bounds its flat block construction separately [8]. The tree packet uses fixed trees and explicitly leaves uniform analysis of input-dependent trees open [10]. Theorem 4.1 controls the complete operation sets, including block construction, under a single adaptive choice. Its proof uses elementary convexity and the inherited entropy inequality. The contribution is the certified composition and resulting bound within the inspected corpus; no literature-wide priority claim is made for averaging, tree portfolios, or fast matrix multiplication.

4.2 The two explicit three-plan certificates

Group labels start at zero. The order within a block matters because it determines the recursive split. Vertical bars separate the final blocks.

Groups	Plans
9	$(0,1,2) \mid (3,4,5) \mid (6,7,8)$ $(6,2,5) \mid (4,7,0) \mid (1,8,3)$ $(5,2,8) \mid (4,6,1) \mid (0,7,3)$
13	$(0,1,2,3,4) \mid (5,6,7,8) \mid (9,10,11,12)$ $(12,10,1,6,4) \mid (9,0,3,8) \mid (5,2,7,11)$ $(0,8,11,12,1) \mid (7,10,4,3) \mid (5,2,6,9)$

r	Masks/plan	Tuples	Q	b_r^2
9	21	$21^3 = 9261$	3304	6608/18225
13	33	$33^3 = 35937$	7074	1572/4225

Thus

$$b_9 = \frac{4\sqrt{413}}{135} = 0.602145227642\ldots, \qquad b_{13} = \frac{2\sqrt{393}}{65} = 0.609976233895\ldots.$$

These are exhaustive integer certificates, not samples of entropy vectors. An independently written checker reconstructs every support from vertex membership in operation tripartitions, then enumerates every support tuple. The sums of all tested squared norms are respectively 18,018,504 and 107,119,584, providing auxiliary reproducible checks. The primary and independent implementations agree. All check conditions remain enabled under `python -O`.

The finite certificate calculation can be specified without any numerical optimization:

```
Q = 0
for v1 in supports(plan1):
    for v2 in supports(plan2):
        for v3 in supports(plan3):
            Q = max(Q, sum((v1[e]+v2[e]+v3[e])**2
                           for e in unordered_pairs))
```

The search that suggested the plans is irrelevant to validity: the fixed plans and their complete exact checks constitute the certificate.

5 A fully paid $9 \to 13 \to 7$ recursion

5.1 Local recurrence

Each group has $2(r-1)$ incident directed pairs. Lemma 3.1 therefore bounds the total number of entries to cache, up to fixed and polynomial factors, by $2^{(\alpha_r+o(1))\mathcal{K}}$, where

$$\alpha_r = \frac{\sqrt{2(r-1)}}{r}.$$

Every entry calls the actual recursive child contraction. Combining this with Theorem 4.1 gives

$$T_d \le \mathrm{poly}(s)\left[2^{(\alpha_{r_d}+o(1))\mathcal{K}_d} T_{d+1} + 2^{(b_{r_d}+o(1))\mathcal{K}_d}\right]. \tag{9}$$

The three-plan coefficients apply at $r = 9, 13$. At $r = 7$, use one fixed plan with blocks $(0,1,2) \mid (3,4) \mid (5,6)$. Its largest squared directed support gives

$$b_7^2 = \frac{632}{1225}, \qquad b_7 = 0.718275\ldots.$$

The common polynomial exponent in (9) is fixed because there are only three possible group counts. Exponential tables are sorted and scanned in their relevant index order; index lengths and record lengths are polynomial. Reshaping and sorting add polynomial factors in those lengths, not an unaccounted exponential random-access assumption.

5.2 Strict rational schedule

Define

$$\begin{aligned}
r_0 &= 9, & \eta_0 &= \frac{63}{4000},\\
r_1 &= 13, & \eta_1 &= \frac{9}{800},\\
r_d &= 7 \quad (d \ge 2), & \eta_d &= \frac{77}{16200}\left(\frac{4}{9}\right)^{d-2},\\
u_d &= \lfloor \eta_d \lambda \rfloor, & B &= \lceil 6 \log_2 \lambda \rceil,\\
a_d &= \sum_{j=0}^{d} (u_j + B), & H &= \left\lceil \frac{s}{\lambda^{3/4}} \right\rceil.
\end{aligned} \tag{10}$$

Use the existing padding construction with relative slack λ^{-3}. The strict polynomial packing margins below permit this choice and give uniform relative padding error $O(\lambda^{-3}) + s^{-c+o(1)}$ for a fixed $c > 0$.

Its product over $O(\log \lambda)$ levels is $1 + o(1)$. Calls with $M \le H$ are leaves. For sufficiently large s, all active u_d are positive; a finite exceptional range is evaluated directly. The exact capacity identities are

$$\eta_0 + \eta_1 = \frac{27}{1000}, \qquad \sum_{d \ge 0} \eta_d = \frac{8}{225}. \tag{11}$$

5.3 Packing and a noncircular depth argument

The packing conditions imported from asymmetric balancing have, up to polylogarithmic factors, the terms $Q^2\Delta$ at the root and QR, Q^2C, 2^aQ^2 later, where $Q = r^a$. Initially both sides have degree at most $s^{9/10+o(1)}$. At later calls the retained side has degree at most that bound and the fresh opposite side has degree at most $2^{a_{d-1}+1}$ [7, 8, 10].

Exact integer inequalities certify

$$9^{100} < 2^{317}, \qquad 13^{1000} < 2^{3701}, \qquad 7^{100} < 2^{281}.$$

They imply the following strict leading exponent gaps:

Stage	Dominant exponent upper bound	Gap below one
Root	$\frac{9}{10} + 2\frac{63}{4000}\frac{317}{100}$	29/200000
Depth one	$\frac{9}{10} + \frac{27}{1000}\frac{3701}{1000}$	73/1000000
Tail	$\frac{9}{10} + \frac{8}{225}\frac{281}{100}$	1/11250

The other fresh-side and packet-capacity terms are bounded by the exponent

$$\frac{8}{225}\left(2\frac{3701}{1000} + 1\right) < 1.$$

All new packet degrees have exponent at most $8/225 + o(1) < 9/10$.

To avoid assuming the depth conclusion while proving it, consider hypothetical active depths up to $3\log_3 \lambda/4 + O(1)$. In the tail,

$$u_d \ge \lambda^{\varepsilon+o(1)}, \qquad \varepsilon = 1 - \frac{3\ln(9/4)}{4\ln 3} > 0.$$

The cumulative guard contributes $O((\log \lambda)^2)$ to a_d. The fixed positive exponent gaps above absorb these guards, $s/M < \lambda^{3/4}$, and all packing polylogarithms uniformly. Consequently padding is nearly lossless throughout the hypothetical range, and every tail child has size $(2/7 + o(1))M < M/3$ eventually. Two fixed prefix levels do not change the depth bound. An active tail cannot reach the hypothetical limit, closing the bootstrap. Floors are uniformly negligible and padding errors accumulate to $1 + o(1)$. In particular,

$$M_0 \le s, \quad M_1 \le (1 + o(1))\frac{2s}{9}, \quad M_d \le (1 + o(1))\frac{4s}{117}\left(\frac{2}{7}\right)^{d-2} \quad (d \ge 2). \tag{12}$$

5.4 Precision, entries, and leaves

Every retained valuation has $i < u_d$, so $D = 2^{a_d - i} \ge 2^{a_{d-1}+B+1}$. The inherited creation-time precision is at most $O(s\lambda^2 2^{a_{d-1}})$. Relative to MD/u_d, it is bounded by

$$O\left(\lambda^{15/4}2^{-B}\right) = o(1).$$

The Hermite evaluation loss is lower order as well. Thus the sharp precision and entropy estimates hold uniformly. All arithmetic bit lengths stay polynomial in s, since $a_d \le (8/225)\lambda + O((\log\lambda)^2)$.

Direct leaf contraction takes $2^{O(s/\lambda^{3/4})}$ local work. Original entry routines take $2^{O(s^{9/10})}$ times a polynomial. Fresh packet routines take $\exp(O(2^{a_d}\log s)) = \exp(s^{8/225+o(1)})$. All are lower order on $s/\sqrt{\lambda}$. Immutable creation-time tensor descriptions are retained, so later entry evaluation does not recursively expand a previous contraction algorithm. The ancestor child-enumeration costs multiplying leaf work are included in the next calculation.

5.5 Exact sum of the recursion

On the scale $s/\sqrt{\lambda}$, put

$$K_0^* = \frac{\kappa}{\sqrt{\eta_0}}, \qquad K_1^* = \frac{2\kappa}{9\sqrt{\eta_1}}, \qquad K_2^* = \frac{4\kappa}{117\sqrt{77/16200}}.$$

For $d \ge 2$, the successive $\mathcal{K}_d$ coefficients decrease by

$$\gamma = \frac{2/7}{\sqrt{4/9}} = \frac{3}{7}.$$

Unrolling (9), including the infinitely extended tail as an upper bound, gives the maximum of

$$\begin{aligned} C_0 &= b_9 K_0^*, \\ C_1 &= \alpha_9 K_0^* + b_{13} K_1^*, \\ C_\infty &= \alpha_9 K_0^* + \alpha_{13} K_1^* + K_2^* \max\left\{ b_7, \frac{\alpha_7}{1-\gamma} \right\}. \end{aligned} \tag{13}$$

Here

$$\frac{\alpha_7}{1-3/7} = \frac{\sqrt{3}}{2} > b_7,$$

so all tail aggregation terms and the terminal leaf route are covered. Exact rational interval calculations give

Branch	Coefficient
C_0	$30.753912832054975\ldots$
C_1	$30.891018920366092\ldots$
C_∞	$30.512927432780513\ldots$

Thus $C_T = C_1$, giving (1). Products of the fixed polynomial overheads across $O(\log\lambda)$ depths add only $O(\lambda\log\lambda)$ to the exponent. Input descriptions are read with a single fixed polynomial dependence on $s + \mathcal{L}$.

For space, local tables are bounded by the selected operation cost, as proved in Theorem 4.1. Ancestor tables on one active recursion path add in size; their exponents do not add. Consequently the space coefficient is

$$\max\{b_9 K_0^*, b_{13} K_1^*, b_7 K_2^*\} = b_9 K_0^* = C_S.$$

This sharper memory argument is necessary. Using the old root boundary cardinality coefficient $2/3$ alone would give about 34.0493, and would miss the benefit of selecting the tree from the actual entropy vector.

Theorem 5.1 (Main normalized contraction certificate)**.** *Under the explicit normalized interface and imported exact identities of Section 3, the deterministic construction above contracts the root network in*

$$(s+\mathcal{L})^{O(1)}2^{(C_T+o(1))s/\sqrt{\lambda}} \quad \textit{time}, \qquad (s+\mathcal{L})^{O(1)}2^{(C_S+o(1))s/\sqrt{\lambda}} \quad \textit{space},$$

where C_T, C_S are (1)–(2). The respective eventual coefficients 30.892 *and* 30.754 *are valid.*

Proof. Exactness follows from the inherited contraction identity (4): cached tables and the selected plan only reassociate finite sums and products. Every weight allocation is included once. The strict schedule validates all imported packing, degree, precision and entry-access premises. The paid recurrence and local storage analysis above give the asserted resources, including leaves and polynomial overhead. □

6 Why this does not remove the square-root logarithm

6.1 Exact boundary width for complete pair graphs

The following theorem concerns a specified representation: constructing complete subset-boundary tables. It is not a lower bound for SAT or all tensor-contraction algorithms.

Theorem 6.1 (Complete directed full-table width)**.** *Give every ordered pair of distinct vertices in $[r]$, $r \geq 3$, an independent index set of size $Q \geq 2$. A table for a subset B retains every crossing pair index, hence has $Q^{2|B|(r-|B|)}$ entries. Among binary full-table contraction trees, the minimum possible maximum boundary-index count is exactly*

$$W_r = 2\left\lceil \frac{r}{3} \right\rceil \left(r - \left\lceil \frac{r}{3} \right\rceil \right) \geq \frac{4r^2}{9}. \tag{14}$$

Proof. Starting from the root, while a visited subset has more than $2r/3$ vertices, descend to its larger child. At the first subset of size at most $2r/3$, its size k is strictly greater than $r/3$. Therefore

$$\lfloor r/3 \rfloor + 1 \leq k \leq \lfloor 2r/3 \rfloor.$$

The minimum of $2k(r-k)$ on this integer interval is W_r, by concavity and checking r modulo three. This proves the lower bound. For the upper bound, partition into three blocks of sizes differing by at most one, contract within each, merge two blocks, and merge the third. Every internal block subset has size at most $\lceil r/3 \rceil$; the merged two-block boundary equals the boundary of the remaining block. All are bounded by W_r. □

An implementation materializing these tables must store at least Q^{W_r} cells, or perform at least that many output operations if it emits every entry without retaining the full table. Adaptive permutations do not alter the symmetric example. Even a hypothetical matrix exponent of two cannot avoid this output-size cost. Sparse, compressed, implicit or cancellation-based representations are outside this theorem's scope.

6.2 A feasible exact-budget symmetric example

One should not simply postulate equal large exact-list cardinalities. Suppose each directed pair has an identical inventory consisting of c copies of each positive weight $w_0, \ldots, w_{J-1}$. Restrict those exponents to zero or one. The 2^{cJ} choices have total weight at most

$$B_* = c \sum_{q<J} w_q.$$

By the pigeonhole principle some exact weight b_* has at least $2^{cJ}/(B_*+1)$ choices. The identical inventories permit the same weight and cardinality on every pair. The allocation is globally feasible whenever $r(r-1)B_* < p$.

For fixed r, take weights $w_q \le A + 4q$ with

$$c = \Theta\left(\frac{M}{r^2 D}\right), \quad J = \Theta\left(\frac{D}{\sqrt{u}}\right), \quad A = o(D/\sqrt{u}), \quad p = \Theta(MD/u).$$

With sufficiently small constants and enough available modes, $r(r-1)B_* < p$, while $cJ = \Theta(M/(r^2\sqrt{u}))$. When this last quantity dominates $\log(p+1)$, the pigeonhole loss is lower order and

$$\log_2 Q = \Omega\left(\frac{M}{r^2\sqrt{u}}\right).$$

Combining with (14) gives explicit-table work $\exp(\Omega(M/\sqrt{u}))$. Since the root taper has $u = O(\log s)$ and $M = \Theta(s)$, merely changing trees or matrix multiplication does not prove $\exp(o(s/\sqrt{\log s}))$ within this full-table inventory model. This is an admissible abstract index-inventory example, not a claim that every SAT input realizes identical inventories with nonzero coefficients.

6.3 What a true order improvement still needs

The reviewed taper and rank analyses already constrain simple retuning [5, 6]. Our width theorem adds a matching obstruction for dense explicit tables. A route around it must construct additional structure, with full costs: an implicit representation of the actual group functions closed under the required contractions; a structured mode-sum algorithm using $\exp(O(M/u))$ admissible recursive calls; or a different deformation/recovery interface that reduces precision or mode multiplicity. This report does not construct those interfaces. In particular, neither faster matrix multiplication nor a better tree-selection heuristic alone establishes the conditional $s/\log s$ contraction bound.

7 Verification, deliverables, and proof status

The mathematical proof is supplemented by reproducible exact checks. The checks validate finite identities, the numerical certificates, and algebraic regrouping; they do not substitute for the uniform asymptotic argument. The archive contains:

- Complete `report.tex` and the compiled `report.pdf`.
- A primary support generator and exact tree selector, the fixed plan data, and an independently implemented certificate checker.
- A rational schedule checker. It bounds π using Machin's identity and alternating arctangent series, bounds $\ln 2$ by its positive atanh series with an explicit tail, and bounds radicals with integer square roots. The decimal thresholds are certified without floating-point tests.
- Independent exact contraction tests over $(\mathbb{Z}/64\mathbb{Z})[V]/(V^2)$, including zero and nilpotent cases, both orientations of a pair, unequal index dimensions, and each adaptive branch. A literal directed-index sum is the oracle.
- A boundary-width dynamic program checking all binary split shapes by cardinality for $3 \le r \le 150$, and exact-budget pigeonhole fixtures.
- Source provenance, a twenty-packet audit, independent mathematical review notes, proof-status metadata, replay outputs, and a checksum manifest.

The finite algebra audit covers 17 fixtures, 43 candidate-tree contractions, 17 independently selected dispatches, 31,681 literal directed-index assignments, and 1,095 checks that operation costs dominate input/output table sizes. All ordinary fixtures have nonzero results, so agreement is not explained by accidental zero products. Deliberate zero and double-nilpotent fixtures remain zero. Normal and optimized Python replays agree, with explicit exception checks enabled in both modes.

Limits of verification. No Lean or Rocq formalization was built in this work. The upstream proof assistant development was not rechecked. No complete SAT solver or Coppersmith–Winograd-type matrix multiplication backend is included. The mathematical dependency boundary is explicit: upstream deformation, integrality, recovery, reduction and packet interfaces are imported; the support certificate, adaptive choice, paid schedule analysis and scoped table-width argument are supplied here. Published or archived benchmark times are not asserted to be reproducible SAT speedups.

7.1 Replay and integration

From the extracted directory, use Python 3.10 or later:

```
python3 verify_manifest.py
python3 run_checks.py
python3 build_report.py
```

On Windows, `python` can replace `python3`. The check programs use the standard library. The report build requires pdfLaTeX and the listed standard packages; shell escape is disabled. Verification runs before regenerating outputs preserve the original manifest check.

The integration point is the exact per-pair aggregation layer of the incoming contraction-tree method. Keep the existing canonical lifts, precision labels and child construction. Count exact pair lists, select one of the three certified plans, build each group cache once, and execute only that plan. Use the strict schedule in (10) and retain the separate PS and special-degree branches where their premises or memory guarantees are desired. The supplied changes are delivered as a standalone research packet; the reviewed repository was not modified.

A Reference implementation contract

The primary selector accepts one positive integer list cardinality for every directed pair. Zero cardinalities mean that an exact-weight allocation contributes zero and must be skipped before selection. The selector compares the integer fifth powers of predicted operation costs. It never computes a floating-point logarithm to choose a branch.

For a merge with product dimensions d_1, d_2, d_3, this fifth power is

$$\frac{(d_1 d_2 d_3)^5}{\min(d_1, d_2, d_3)^3}.$$

The division is exact. Equivalently it is the maximum of the three products assigning exponents $2, 5, 5$ to the dimensions. The complete plan's score is the maximum over its operations. The exact certificate bounds the selected score over all admissible entropy vectors through weighted supports, without assuming that small toy fixtures represent all asymptotic inputs.

B Exact constants for the independent schedule checker

The checker represents every coefficient as $\pi/\ln 2$ times a sum of square roots of rational numbers:

$$C_0 = \frac{\pi}{\ln 2}\sqrt{\frac{302080}{6561}},$$
$$C_1 = \frac{\pi}{\ln 2}\left(\sqrt{\frac{128000}{5103}} + \sqrt{\frac{134144}{41067}}\right),$$
$$C_\infty = \frac{\pi}{\ln 2}\left(\sqrt{\frac{128000}{5103}} + \sqrt{\frac{51200}{41067}} + \sqrt{\frac{4800}{13013}}\right).$$

It verifies $C_1 > C_0, C_\infty$, $C_1 < 30.892$, and $C_0 < 30.754$. The same checker pays for all later local storage terms. The local adaptive squares 6608/18225 and 1572/4225 are separate inputs whose exact finite certificates are verified by the independently reconstructed operation sets.

References

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[2] OpenAI, *The Exponential Time Hypothesis is False*, October 25, 2026. Preprint directory at upstream commit 7ef5c0cc5c. The contraction and quantitative reduction appendices are the principal imported interface. An unmodified copy is vendored in ProveIt.

[3] *A $2^{o(n)}$ time algorithm for 3-SAT*, credited to Claude (Anthropic), October 26, 2026, 49-page manuscript, archived PDF.

[4] *Density-aware subexponential SAT improvement* and *ETH density refinement*, October 26, 2026, first packet; second packet.

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[6] *3SAT round-two refinement*, actual-domain rank gate and alternate-manuscript audit, packet source.

[7] *ETH asymmetric packet balancing*, packet source.

[8] *Adaptive aggregation for subexponential SAT*, October 26, 2026, `sat_adaptive_contraction_2026-10-26.zip`, immutable archive.

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[11] J. Alman, R. Duan, V. Vassilevska Williams, Y. Xu, Z. Xu, and R. Zhou, *More Asymmetry Yields Faster Matrix Multiplication*, arXiv:2404.16349, version 2 (2024), `https://arxiv.org/html/2404.16349v2`. Only the strict inequality $\omega < 12/5$ is needed; no current-record claim or newer numerical optimization is required.